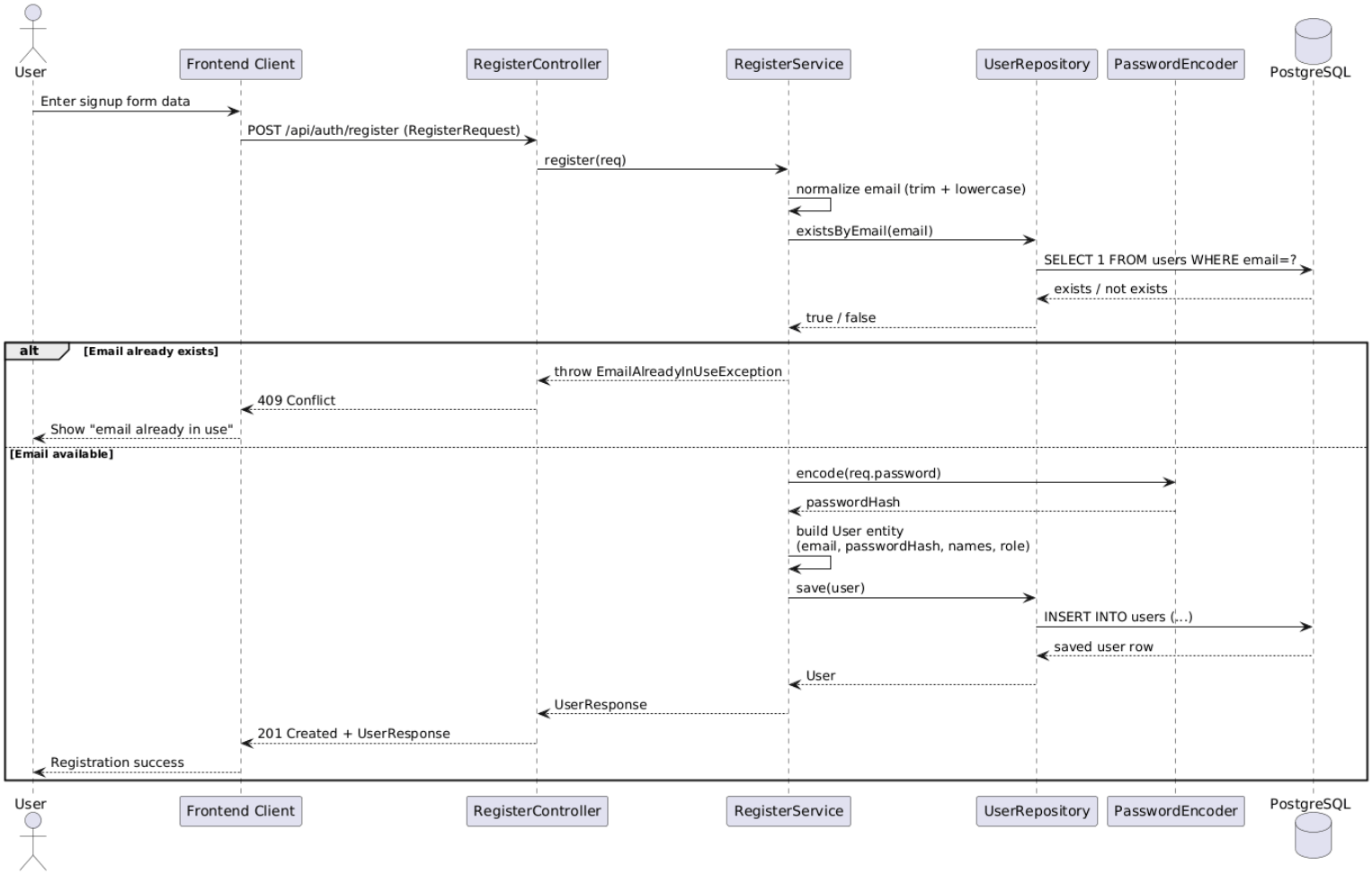
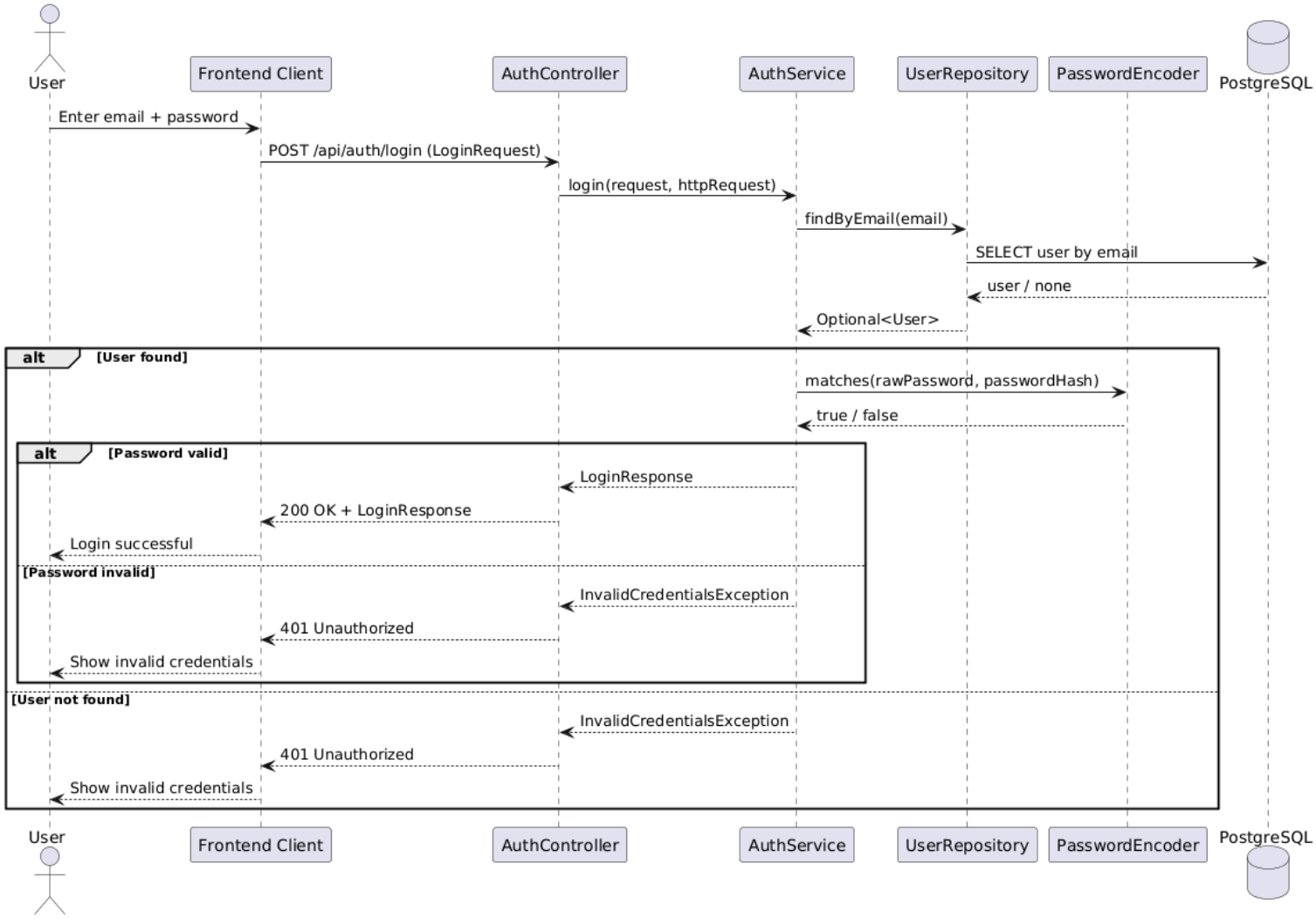
Feature 1: User authentication & Role management

Register:

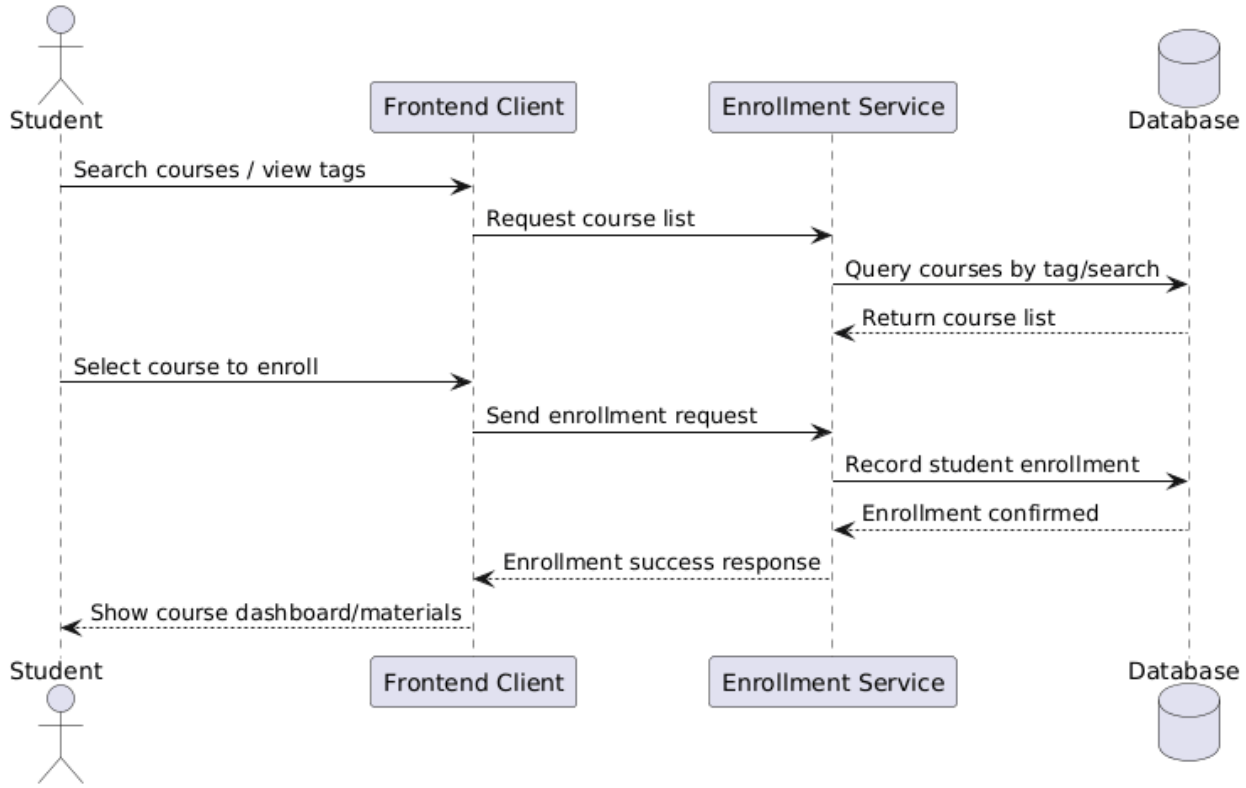
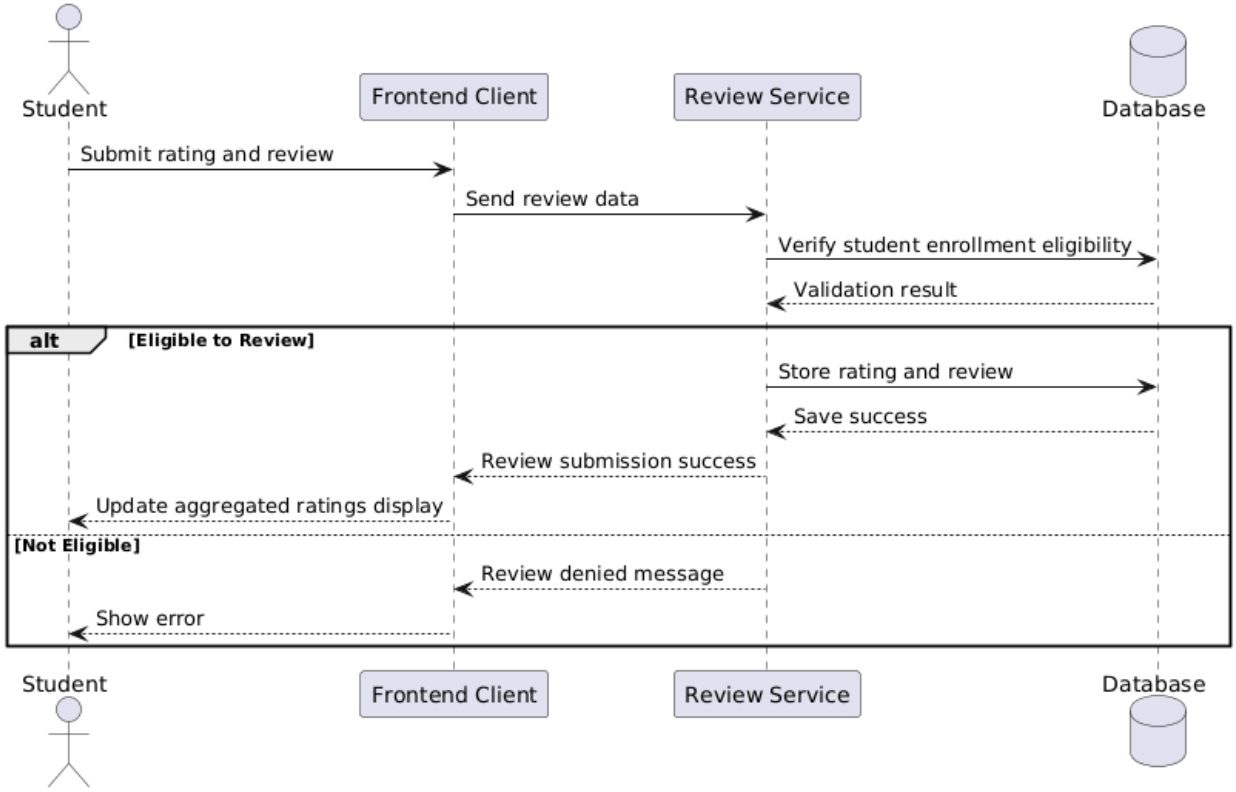


Login:


User
Frontend Client
AuthController
AuthService
UserRepository
PasswordEncoder
PostgreSQL
Enter email + password
POST /api/auth/login (LoginRequest)
login(request, httpRequest)
findByEmail(email)
SELECT user by email
user / none
Optional<User>
alt
[User found]
matches(rawPassword, passwordHash)
true / false
alt
[Password valid]
LoginResponse
200 OK + LoginResponse
Login successful
[Password invalid]
InvalidCredentialsException
401 Unauthorized
Show invalid credentials
[User not found]
InvalidCredentialsException
401 Unauthorized
Show invalid credentials
User
Frontend Client
AuthController
AuthService
UserRepository
PasswordEncoder
PostgreSQL

Feature 2: Course Creation & Management

Feature 3: Enrollment & Progress Tracking



Feature 4: Enrollment & Progress Tracking